[INCH-POUND]
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COMMERCIAL ITEM DESCRIPTION

MILLING MACHINE, COMBINATION HORIZONTAL AND VERTICAL, KNEE TYPE

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers horizontal and vertical spindle, knee milling machines intended for use in conventional milling operations.

2. CLASSIFICATION. The milling machines shall be of the following classes and sizes. The class and size to be furnished shall be as specified (see 7.2(b)).

Class 1 - 5-horsepower horizontal and 4-horsepower vertical spindle drives
Class 2 - 7.5-horsepower horizontal and 4-horsepower vertical spindle drives
Class 3 - 7.5-horsepower horizontal and 5-horsepower vertical spindle drives
Class 4 - 10-horsepower horizontal and 4-horsepower vertical spindle drives
Class 5 - 20-horsepower horizontal and 15-horsepower vertical spindle drives

Size - Sizes are shown in table I

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Defense Supply Center Richmond, ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3417

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TABLE I. Milling machines sizes.

Size	Class	Minimum machine travel, inches		
		Longitudinal	Cross	Vertical
1	1	29	11	18
2		33	11	18
3		36	16	17
4	2	29	12	17
5		35	14	17
6		41	13	18
7	3	29	12	17
8		35	14	17
9		41	13	18
10	4	31	12	16
11	5	52	14	24

Note: The range of machine sizes shown in table I shall be equal to or larger than the size stated but smaller than the next available size. When the largest machine size is ordered, the machine offered shall not exceed the stated requirements by more than 10 percent.

3. SALIENT CHARACTERISTICS

3.1 General requirements. The milling machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. The range of machine sizes shown in table I shall be equal to or larger than the size stated but smaller than the next available size. When the largest machine size is ordered, the machine offered shall not exceed the stated requirements by more than 10 percent.

3.2 Components. The milling machine shall include, as a minimum, the following components:

3.2.1 Column and base. The milling machine column and base shall support the other elements of the machine, provide stability for the table, minimize adverse vibration, and provide a means for leveling and anchoring the machine.

3.2.2 Knee. The knee shall support the worktable and provide for clamping in any position within its travel on the column. The knee adjusting crank or handwheel shall positively and automatically release when the power feed or rapid traverse is engaged.

3.2.3 Worktable. Unless otherwise specified (see 7.2(c)), the worktable length, width, and weight-holding capacity shall be that considered standard by the manufacturer. The number and size of the T-slots shall be as specified (see 7.2(d)). The configuration of the T-slots shall conform to the requirements of American National Standards Institute (ANSI) B5.1, T-slots, Their Bolts, Nuts, Tongues, and Cutters. When specified (see 7.2(e)), the worktable shall rotate ±45 degrees left and right of center.

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3.2.4 Overarm. The overarm shall be mounted on top of the column and shall allow for support of the vertical spindle and horizontal cutter arbor. When specified (see 7.2(f)), the overarm shall be mounted on a turret that shall allow the overarm to swivel 180 degrees in a horizontal plane.

3.2.5 Horizontal spindle. Unless otherwise specified (see 7.2(g)), the horizontal spindle shall be capable of 10 speeds ranging from 55 to 1600 revolutions per minute. The taper size of the spindle shall be as specified (see 7.2(h)) and shall be in accordance with ANSI B5.18, Spindle Noses and Arbors for Milling Machines.

3.2.6 Vertical spindle. Unless otherwise specified (see 7.2(i)), the vertical spindle shall be capable of 10 speeds ranging from 120 to 1400 revolutions per minute. The taper size of the spindle shall be as specified (see 7.2(j)) and shall be in accordance with ANSI B5.18. If required, the vertical spindle quill travel shall be as specified (see 7.2 (k)).

3.2.7 Digital readout. Unless otherwise specified (see 7.2(l)), a digital readout shall be provided for positional readout of all axes. The resolution and accuracy shall be within ±0.0005 inch. The repeatability shall be within ±0.0002 inch measured over the full travel on each axis. The digital readout shall indicate axis location in both English and metric units.

3.2.8 Universal dividing head. When specified (see 7.2(m)), a universal dividing head shall be furnished. The universal dividing head shall index and rotate workpieces in conjunction with the longitudinal movement of the worktable. The index plates shall be those normally furnished with the standard model dividing head. When specified (see 7.2(n)), additional index plates shall be furnished for dividing a single spindle rotation into any integral number of equal divisions from two to 100 when using the plain indexing method. When specified (see 7.2(o)), a complete driving mechanism with enclosed gears shall be provided to drive the dividing head by power from the worktable feed screw. The dividing head shall be complete with centers, a steady rest, and a three-jaw universal chuck with reversible jaws.

3.2.9 Rotary table. When specified (see 7.2(p)), a rotary table shall be provided complete with power drive system, handwheel, two trip dogs, and locking clamp. Unless otherwise specified (see 7.2(q)), the rotary table shall have a peripheral angular scale divided into one-degree increments. The table diameter and weight-holding capacity shall be as specified (see 7.2(r)). If required, an indexing attachment and index plates shall be furnished as specified (see 7.2(s)).

3.2.10 Controls. The milling machine shall be equipped with worktable, saddle, and knee manual and power feed. Unless otherwise specified (see 7.2(t)), the power feed systems shall include a means to eliminate backlash. The milling machine shall be equipped with powered rapid traverse for positioning the workpiece.

3.2.10.1 Manual feed. The machine shall be equipped with cranks or handwheels to move the worktable, saddle, and knee. The crank or handwheel scale shall be graduated in 0.001-inch increments.

3.2.10.2 Power feed. Worktable, saddle, and knee power feeds shall be engaged by individual controls on the worktable and knee. When specified (see 7.2(u)), an adjustable control pendant with power feed controls shall be provided. Axis travel limit safety stops shall be provided to automatically disengage the power feed to prevent accidental damage to the feed mechanism. Adjustable power feed

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trip dogs shall be provided to disengage the feed at preset points. Unless otherwise specified (see 7.2(v)), the milling machine shall have longitudinal, cross, and vertical axis step feed rates considered by the manufacturer as standard. When specified (see 7.2(w)), the milling machine's longitudinal, cross, and vertical feed rates shall be continuously variable over the speed range specified.

3.2.10.3 Rapid traverse. When engaged, the rapid traverse shall override the feed rate of the worktable, saddle, or knee in the direction of the feed. Unless otherwise specified (see 7.2(x)), the longitudinal, cross, and vertical axis rapid traverse rates shall be not less than 72 inches per minute, 72 inches per minute, and 20 inches per minute, respectively.

3.2.10.4 Spindle controls. The milling machine shall be equipped with spindle controls including main motor forward and reverse switch(s), clutch(s), and spindle speed control(s).

3.2.10.5 Locks. The milling machine shall be equipped with locks on the worktable, saddle, and knee to permit the components to be locked in place during machining operations.

3.2.10.6 Automatic cycle worktable control. When specified (see 7.2(y)), the milling machine shall have an automatic worktable control to provide automatic work cycles by controlling power feed, automatic reverse, rapid traverse, and stop.

3.2.11 Coolant system. Unless otherwise specified (see 7.2(z)), the coolant system shall be the flood type. The coolant system shall provide a means for directing and controlling the rate of coolant flow. The coolant returning to the central reservoir shall be screened of particles exceeding 1000 microns.

3.2.12 Electrical system. Unless otherwise specified (see 7.2(aa)), the electrical system shall conform to the National Fire Protection Association (NFPA) 79, Electrical Standards for Industrial Machinery. The machine shall operate from a dual 230/460 volt, 3-phase, 60-hertz circuit and shall be wired for the voltage specified (see 7.2(bb)). Unless otherwise specified (see 7.2(cc)), control system power shall be provided by a separate 120 volt, single-phase, 60-hertz circuit from the machine's main power supply.

3.2.13 Spindle drive motors. The spindle drive motors shall have a 30-minute duty cycle horsepower rating. The spindle drive motor shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturers Association (NEMA) MG 1, Motors and Generators. Unless otherwise specified (see 7.2(dd)), the motors shall have open drip-proof enclosures.

3.2.14 Optional accessories. Optional accessories shall be furnished as specified (see 7.2(ee)).

3.3 Performance. Unless otherwise specified (see 7.2(ff)), the milling machine shall meet the alignment requirements shown in table II when tested in accordance with Aerospace Industries Association (AIA) National Aerospace Standard (NAS) 985, Uniform Alignment Tests – NAS 900 Series Equipment Specifications.

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TABLE II. Alignment requirements.

NAS 985 paragraph number	Test parameter
4.3.2.1	Flatness – work-mounting surface - Maximum deviation in any 12 inches: 0.0008 inch - Maximum overall accumulated error shall not exceed 0.0008 inch plus 0.00015 inch per foot for either length or width
4.3.3.1	Rise and fall of work-mounting surface - Maximum deviation in any 12 inches: 0.0008 inch - Maximum overall accumulated error shall not exceed 0.0008 inch plus 0.00015 inch per foot for either length or width
4.3.4.1	T-slots parallel with longitudinal movement - Maximum deviation in any 12 inches: 0.0008 inch - Maximum overall accumulated error shall not exceed 0.0008 inch plus 0.00015 inch per foot for total T-slot length
4.3.5.1	Transverse movement square with center T-slot - Maximum deviation in any 12 inches: 0.0008 inch - Maximum overall accumulated error shall not exceed 0.0008 inch plus 0.00015 inch per foot for total travel up to 0.003 inch
4.3.6.1	Center T-slot square with spindle - Maximum deviation shall not exceed 0.001 inch in 18-inch sweep - Compensate for any error indicated in test section 4.3.4.1
4.3.8.1	Work-mounting surface square with spindle (vertical) - 12-inch sweep 0.001 inch TIR - 24-inch sweep 0.002 inch TIR
4.3.9.1	Transverse movement square with longitudinal movement - Maximum deviation in any 12 inches: 0.0008 inch - Maximum overall accumulated error shall not exceed 0.0008 inch plus 0.00015 inch per foot for total travel up to 0.003 inch
4.3.10.1	Spindle runout - Radial runout: 0.0003 inch TIR - Axial (face) runout: 0.0002 inch TIR - 1.25 inches from face: 0.0003 inch TIR - 12 inches from face: 0.001 inch TIR
4.3.13.1	Alignment of spindle with arbor support - Runout shall not exceed 0.001 inch TIR with arbor support clamped
4.3.14.1	Spindle parallel with overarm - Maximum deviation shall not exceed 0.002 inch TIR overall
4.3.15.1	Saddle transverse movement parallel with spindle - Maximum deviation shall not exceed 0.00075 inch per foot
4.3.17.1	Spindle parallel with work-mounting surface - Maximum deviation shall not exceed 0.001 inch per foot with worktable in unclamped position
4.3.18.1	Work-mounting surface square with vertical knee movement - Maximum deviation shall not exceed 0.005 inch per foot
4.3.21.1	Deflection of worktable under load - Worktable shall support 350 pounds per square foot on overhanging portion with a maximum deflection of 0.015 inch
4.3.22.1	Slide displacement due to clamping - Movement of knee assembly when clamp is locked shall not exceed 0.001 inch

Note: TIR – Total Indicator Reading

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3.4 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and machine weight shall not exceed the restrictions specified (see 7.2(gg)).

3.5 Safety and health requirements. The machine shall be designed and manufactured in accordance with ANSI B11.8, Safety Requirements for Construction, Care, and Use of Drilling, Milling, and Boring Machines and Machine Tools. The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance with Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910. If a conflict arises between the ANSI and OSHA standards, the OSHA standards shall apply.

3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(hh)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2(ii)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.7 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2(jj)), the nameplate shall contain the following information:

a. Nomenclature.
b. Manufacturer's name.
c. Serial number.
d. Machine model designation.
e. Power input (volts, total amperes, phase, frequency).
f. Short-circuit/over-current rating.
g. Contract number or order number.
h. National stock number.
i. Date of manufacture.

3.8 Lubrication plate or chart. When specified (see 7.2(kk)), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application.
b. Servicing interval.
c. Type of lubricant(s) with SAE number or lubricant identifier.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

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5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description, conform to the manufacturer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Inspection. The milling machine shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2(ll)), the milling machine shall be tested in accordance with sections 5.3.1 through 5.3.4.

5.3.1 Test setup and conditions. Unless otherwise specified (see 7.2 (mm)), the manufacturer shall be responsible for supplying all tooling and materials and shall conduct all tests required by the Government. The manufacturer shall provide test instruments and equipment with calibration traceable to National Institute of Standards and Technology (NIST) and shall provide certification of calibration as specified (see 7.2(nn)). Unless otherwise specified (see 7.2(oo)), the milling machine shall be tested in the standard environmental conditions of the manufacturer's facility.

5.3.2 Operational test. The milling machine shall be operated in accordance with the manufacturer's standard operating procedures for warm-up and operation. Proper operation of all controls and features shall be verified during the trial period.

5.3.3 Performance tests. The alignment and cutting performance tests shall be conducted in accordance with NAS 985 and NAS 979, Uniform Cutting Tests - NAS Series Metal Cutting Equipment Specifications.

5.3.3.1 Alignment tests. The alignment test results shall meet the requirements shown in table II.

5.3.3.2 Maximum torque cutting test. The horizontal and vertical spindles of the machine shall meet the performance requirements of tests I and II of NAS 979 paragraph 4.3.3.1.1. Tests I and II shall be performed on both the horizontal and vertical spindles using an 8 inch width of cut, a 27.5 inch per minute feed rate, and a type VIII cutter described in table I of NAS 979. Test I shall be performed on the horizontal spindle using a 55 revolutions per minute spindle speed and a 0.100 inch depth of cut. Test II shall be performed on the horizontal spindle using a 55 revolutions per minute spindle speed and a 0.050 inch depth of cut. Test I shall be performed on the vertical spindle using a 112 revolutions per minute spindle speed and a 0.204 inch depth of cut. Test II shall be performed on the vertical spindle using a 112 revolutions per minute spindle speed and a 0.102 inch depth of cut.

5.3.3.3 Maximum rated horsepower cutting test. The horizontal and vertical spindles of the machine shall meet the performance requirements of NAS 979 paragraph 4.3.3.2.2. The test shall be performed using an 12 inch width of cut, a 5.5 inch per minute feed rate, a 112 revolutions per minute spindle speed, a 0.0035 inch per tooth chip load, and a type VI cutter described in table I

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of NAS 979. Spindles of 4, 5, 7.5, 10, 15, and 20 horsepower shall be tested using a depth of cut of 0.044, 0.055, 0.082, 0.109, 0.164, and 0.218 inch, respectively.

5.3.3.4 Maximum feed rate cutting test. The horizontal and vertical spindles of the machine shall meet the performance requirements of NAS 979, paragraph 4.3.3.3.1. The test shall be performed using an 12 inch width of cut, a 25 inch per minute feed rate, a 1200 revolutions per minute spindle speed, a 0.0052 inch per tooth chip load, and a type III cutter described in NAS 979 table I. Spindles of 4, 5, 7.5, 10, 15, and 20 horsepower shall be tested using a depth of cut of 0.058, 0.073, 0.109, 0.145, 0.218, and 0.290 inch, respectively.

5.3.3.5 Flatness and mismatch cutting test. The machine shall meet the performance requirements of NAS 979 paragraph 4.3.3.12.2.

5.3.4 Optional tests. Optional tests shall be conducted as specified (see 7.2(pp)).

5.4 Acceptance. Unless otherwise specified (see 7.2(qq)), preliminary and final acceptance tests shall be conducted at the manufacturer's site. Failure of the machine to meet the performance requirements of section 5.3.3 and the alignment requirements shown in table II shall be cause for rejection.

6. PACKAGING. Unless otherwise specified (see 7.2(rr)), preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES

7.1 Sources of documents.

7.1.1 Government documents. Copies of Federal documents may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

7.1.2 Industry standards. Copies of industry standards referenced in this CID may be obtained from the following addresses:

Aerospace Industries Association (AIA)

NAS 979	Uniform Cutting Tests - NAS Series Metal Cutting Equipment Specifications
NAS 985	Uniform Alignment Test - NAS 900 Series Equipment Specifications

Applications for copies should be sent to the Aerospace Industries Association, 1250 Eye Street NW, Suite 1200 Washington, DC 20005-3924.
American National Standards Institute (ANSI)

ANSI B5.1	T-slots, Their Bolts, Nuts, Tongues, and Cutters

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ANSI B5.18 Spindle Noses and Arbors for Milling Machines

ANSI B11.8 Safety Requirements for Construction, Care, and Use of Drilling, Milling, and Boring Machines and Machine Tools.

Applications for copies should be sent to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

National Electrical Manufacturers Association (NEMA)

NEMA MG 1 Motors and Generators

Applications for copies should be sent to the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

National Fire Protection Association (NFPA)

ANSI/NFPA 79 Electrical Standard for Industrial Machinery

Applications for copies should be sent to the National Fire Protection Association, One Batterymarch Park, P.O. Box 9101, Quincy, MA 02269-9101.

Occupational Safety and Health Association (OSHA)

OSHA 1910 Occupational Safety and Health Standards for General Industry (29 CFR Part 1910)

Applications for copies should be sent to the U.S. Department of Labor, 200 Constitution Avenue NW, Room 423, Washington, DC 20210.

7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Class and size of milling machine required (see 2).
c. Worktable length, width, and weight-holding capacity, if different (see 3.2.3).
d. T-slot number and size, as specified (see 3.2.3).
e. Worktable rotation, if required (see 3.2.3).
f. Turret-mounted overarm, if required (see 3.2.4).
g. Horizontal spindle speed range and number of spindle speeds, if different (see 3.2.5).
h. Horizontal spindle taper, as specified (see 3.2.5).
i. Vertical spindle speed range and number of spindle speeds, if different (see 3.2.6).
j. Vertical spindle taper, as specified (see 3.2.6).
k. Vertical spindle quill travel, as specified (see 3.2.6).
l. Digital readout, if different (see 3.2.7).
m. Universal dividing head, if required (see 3.2.8).

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n. Additional index plates, if required (see 3.2.8).
o. Dividing head drive mechanism, if required (see 3.2.8).
p. Rotary table, if required (see 3.2.9).
q. Rotary table peripheral scale angular increments, if different (see 3.2.9).
r. Rotary table diameter and weight-holding capacity, as specified (see 3.2.9).
s. Rotary table indexing attachment and index plates, if required (see 3.2.9).
t. Power feed systems, if different (see 3.2.10).
u. Adjustable control pendant, if required (see 3.2.10.2).
v. Step feed rates, if different (see 3.2.10.2).
w. Infinitely variable feed rates, if required (see 3.2.10.2).
x. Rapid traverse rates, if different (see 3.2.10.3).
y. Automatic worktable cycle control, if required (see 3.2.10.6).
z. Coolant system, if different (see 3.2.11).
aa. Electrical system industrial standard, if different (see 3.2.12).
bb. Electrical system voltage, as specified (see 3.2.12).
cc. Control system voltage, if different (see 3.2.12).
dd. Motor enclosures, if different (see 3.2.13).
ee. Optional accessories, as specified (see 3.2.14).
ff. Performance, if different (see 3.3).
gg. Machine dimensions and weight, as specified (see 3.4).
hh. System of units required (see 3.6).
ii. Dual system of units, if required (see 3.6).
jj. Nameplate, if different (see 3.7).
kk. Lubrication plate or chart, if required (see 3.8).
ll. Testing, if different (see 5.3).
mm. Test responsibility, if different (see 5.3.1).
nn. Certification of calibration, as specified (see 5.3.1).
oo. Test environmental conditions, if different (see 5.3.1).
pp. Optional tests, as specified (see 5.3.4).
qq. Acceptance, if different (see 5.4).
rr. Packaging requirements, if different (see 6).

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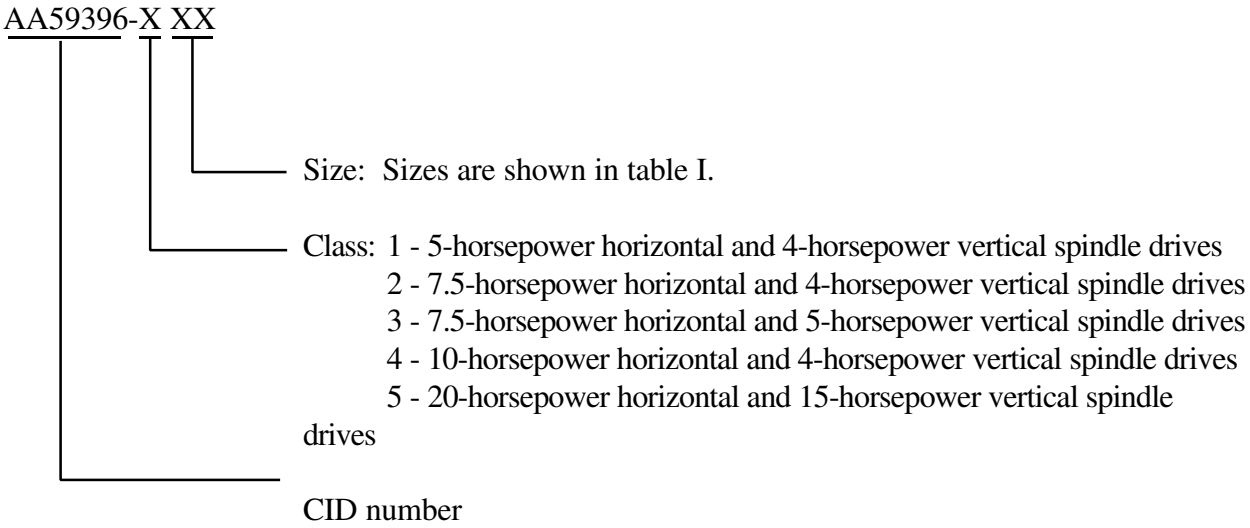
7.3 Part identification number (PIN). The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.



MILITARY INTERESTS:

Custodians
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Army - AL
Navy - SH

Reviewers
Air Force - 84
Navy - MC, YD1

PREPARING ACTIVITY
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